

VILLACH STADT, Austria

WMO: 112130

Lat: 46.618N

Lon: 13.874E

Elev: 1622

StdP: 13.85

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	15.3	6.9	8.3	18.3	10.5	10.0	22.1	13.2	39.9	11.6	40.6	1.6	230	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	87.7	68.8	84.5	67.8	81.5	66.4	70.6	84.8	69.0	81.8	67.5	79.0	5.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	101.4	75.9	64.5	96.6	74.1	63.3	92.5	72.5	35.4	84.8	34.1	82.0	32.8	79.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.4	8.8	DB	7.6	93.0	5.7	3.6	3.5	95.6	0.1	97.7	-3.1	99.7	-7.3	102.3	(4)
(5)			WB	6.9	73.4	5.5	1.5	3.0	74.5	-0.3	75.3	-3.3	76.2	-7.3	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	29.5	34.2	42.2	51.1	59.2	65.9	69.2	68.1	59.3	51.0	40.5	31.0		
	DBStd	15.28	6.72	6.41	7.06	6.74	6.19	6.47	5.69	5.68	5.84	6.79	6.88	6.18		
	HDD50	2356	636	443	252	67	5	0	0	0	3	71	289	590		
	HDD65	5819	1101	863	706	416	198	67	25	33	186	435	734	1055		
	CDD50	2426	0	0	11	101	291	478	595	560	282	102	6	0		
	CDD65	410	0	0	0	0	19	95	154	127	15	0	0	0		
	CDH74	3380	0	0	0	16	199	817	1286	968	92	2	0	0		
	CDH80	982	0	0	0	1	30	234	421	290	6	0	0	0		
(14) Wind	WSAvg	3.3	2.7	2.9	3.6	3.9	3.9	3.8	4.0	3.6	3.0	3.0	2.9	2.6		
(15) Precipitation	PrecAvg	50.6	2.0	2.0	2.8	3.5	4.1	5.2	5.9	5.8	5.4	5.5	5.5	3.0		
	PrecMax	66.6	8.7	9.1	6.7	5.9	7.8	8.9	11.6	8.7	9.3	13.5	18.0	9.2		
	PrecMin	39.1	0.1	0.0	0.1	0.8	1.5	1.9	1.8	2.2	2.2	0.0	0.6	0.4		
	PrecStd	7.3	1.7	2.0	1.7	1.5	1.4	1.7	2.0	1.9	2.0	3.0	3.8	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.6	56.6	67.9	76.3	84.2	90.0	92.7	91.7	80.4	73.1	60.7	48.4		
		MCWB	43.4	45.8	51.6	58.1	65.8	69.5	71.6	70.3	66.3	60.7	51.8	43.6		
	2%	DB	44.7	51.1	63.0	71.3	79.1	85.9	87.9	86.7	76.7	68.3	55.9	44.4		
		MCWB	39.1	42.0	49.5	55.5	62.6	68.3	69.3	68.7	64.1	58.3	50.1	40.9		
	5%	DB	41.2	47.5	58.7	67.9	75.5	82.4	84.6	82.9	73.0	64.8	52.6	40.9		
		MCWB	37.1	39.9	47.4	53.6	60.4	66.3	67.9	67.7	62.0	56.5	48.9	38.1		
	10%	DB	38.3	44.0	54.7	64.1	72.0	78.9	81.3	79.5	69.6	61.2	49.9	38.6		
		MCWB	35.4	38.0	45.2	51.9	58.7	64.7	66.6	66.2	60.2	54.8	47.2	36.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	46.6	52.6	58.9	66.8	71.8	72.9	72.1	67.3	61.5	53.5	44.8		
		MCDB	49.2	55.4	65.3	74.4	80.8	86.3	89.4	87.7	78.1	70.5	56.7	47.0		
	2%	WB	40.3	43.0	50.1	56.6	63.8	69.6	70.9	70.0	65.2	59.4	51.4	41.5		
		MCDB	43.7	49.5	61.7	68.9	76.5	83.1	85.1	83.7	74.5	66.4	54.5	43.7		
	5%	WB	37.7	40.6	48.1	54.7	61.6	67.6	69.1	68.5	63.3	57.5	49.4	38.8		
		MCDB	40.5	46.4	57.7	65.2	73.0	79.6	81.9	80.7	70.9	63.2	52.3	40.5		
	10%	WB	35.8	38.5	46.1	52.9	59.8	65.8	67.4	67.0	61.5	55.7	47.5	36.8		
		MCDB	37.8	43.3	53.4	62.0	69.8	76.3	78.8	77.6	67.8	60.6	49.7	38.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	15.7	19.3	19.0	19.4	19.8	20.5	19.0	17.3	15.5	10.6	10.1		
		MCDBR	16.2	22.1	26.8	26.1	25.7	25.6	25.3	24.8	21.3	20.2	15.0	13.6		
	5% WB	MCWBR	10.8	13.3	14.4	12.2	10.7	10.0	9.3	9.6	10.0	10.8	9.5	9.8		
		MCWBR	13.8	20.4	24.9	23.0	23.6	24.0	23.4	22.9	19.5	18.1	12.6	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.329	0.381	0.432	0.442	0.451	0.447	0.440	0.408	0.381	0.340	0.300		
	taud		2.379	2.300	2.220	2.136	2.165	2.192	2.213	2.247	2.307	2.348	2.410	2.440		
	Ebn at Noon		252	264	265	261	262	260	259	256	253	242	233	239		
	Edh at Noon		23	30	38	45	46	45	43	40	35	29	23	19		
(44) All-Sky Solar Radiation	RadAvg		443	711	1068	1342	1589	1780	1804	1583	1164	773	442	371		
	RadStd		53	99	133	169	180	136	132	162	142	90	78	52		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (89 neighbors)	+0.86	N/A	N/A	+1.30	+0.76	+0.59	N/A	-223	+158

Nomenclature: See separate page